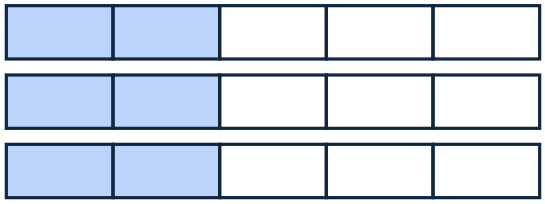
Fractions Times Whole Numbers

3 × 2/5 means 3 groups of 2/5.

3 groups of 2/5



$$3 \times \frac{2}{5} = \frac{6}{5}$$

2 fifths, 3 times: 6 fifths.

It's the same as adding

$$\frac{2}{5} + \frac{2}{5} + \frac{2}{5} = \frac{6}{5} = 1\,\frac{1}{5}$$

Every fraction is unit fractions put together

$$\frac{5}{4} = 5 \times \frac{1}{4}$$

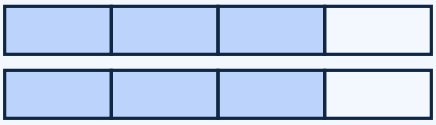
5 pieces, each one 1/4.

Multiply the top. The bottom stays.

$$4 \times \frac{3}{8} = \frac{12}{8}$$

4 × 3 = 12 eighths. The pieces are still eighths.

Your turn!



$$2 \times \frac{3}{4} = \frac{6}{4}$$

For teachers

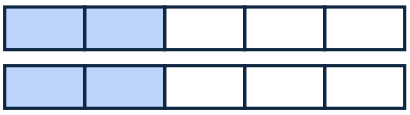
Goal: a whole number times a fraction (4.NF.4): 3 × 2/5 is 3 groups of 2/5. Say “How many groups? How many pieces in each? What size are the pieces?” Only the top number grows.

Fractions Times Whole Numbers: Practice

Groups of a fraction: multiply the top. The bottom stays.

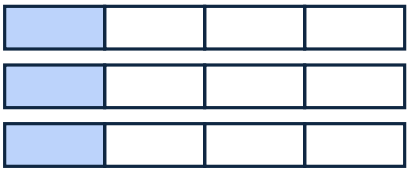
Part 1 Groups of a fraction Count all the shaded pieces.

1)



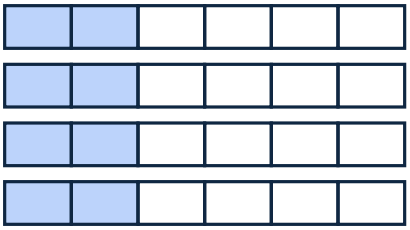
$2 \times \frac{2}{5} = \frac{4}{5}$

2)



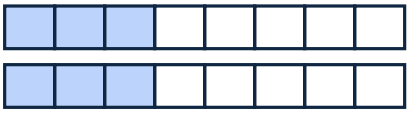
$3 \times \frac{1}{4} = \frac{3}{4}$

3)



$4 \times \frac{2}{6} = \frac{8}{6}$

4)

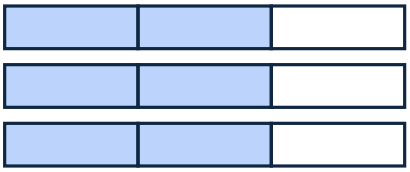


$2 \times \frac{3}{8} = \frac{6}{8}$

Fractions Times Whole Numbers: Practice (continued)

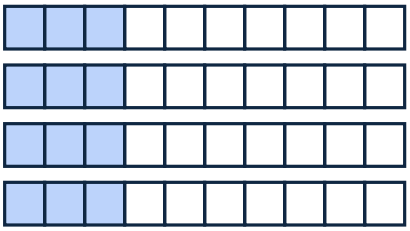
Part 1 Groups of a fraction (continued)

5)



$3 \times \frac{2}{3} = \frac{6}{3}$

6)



$4 \times \frac{3}{10} = \frac{12}{10}$

Part 2 Unit fractions No help this time.

7) $5 \times \frac{1}{4} = \frac{5}{4}$

8) $3 \times \frac{1}{8} = \frac{3}{8}$

9) $7 \times \frac{1}{10} = \frac{7}{10}$

10) $2 \times \frac{1}{3} = \frac{2}{3}$

11) $6 \times \frac{1}{5} = \frac{6}{5}$

12) $4 \times \frac{1}{12} = \frac{4}{12}$

13) $9 \times \frac{1}{6} = \frac{9}{6}$

14) $3 \times \frac{1}{2} = \frac{3}{2}$

15) $11 \times \frac{1}{8} = \frac{11}{8}$

Fractions Times Whole Numbers: Practice (continued)

Part 2 Unit fractions (continued)

16) $\frac{7}{3} = \boxed{7} \times \frac{1}{3}$

17) $\frac{5}{6} = \boxed{5} \times \frac{1}{6}$

18) $\frac{9}{10} = \boxed{9} \times \frac{1}{10}$

19) $\frac{4}{5} = \boxed{4} \times \frac{1}{5}$

20) $\frac{11}{12} = \boxed{11} \times \frac{1}{12}$

21) $\frac{3}{4} = \boxed{3} \times \frac{1}{4}$

Part 3 Add it up Write it as adding, then multiply.

22) $3 \times \frac{2}{6} = \frac{2}{6} + \frac{2}{6} + \frac{2}{6} = \frac{\boxed{6}}{6}$

23) $4 \times \frac{3}{5} = \frac{3}{5} + \frac{3}{5} + \frac{3}{5} + \frac{3}{5} = \frac{\boxed{12}}{5}$

24) $2 \times \frac{5}{8} = \frac{5}{8} + \frac{5}{8} = \frac{\boxed{10}}{8}$

25) $3 \times \frac{4}{10} = \frac{4}{10} + \frac{4}{10} + \frac{4}{10} = \frac{\boxed{12}}{10}$

26) $5 \times \frac{2}{12} = \frac{2}{12} + \frac{2}{12} + \frac{2}{12} + \frac{2}{12} + \frac{2}{12} = \frac{\boxed{10}}{12}$

27) $2 \times \frac{2}{3} = \frac{2}{3} + \frac{2}{3} = \frac{\boxed{4}}{3}$

Fractions Times Whole Numbers: Practice (continued)

Part 4 Multiply No help this time.

28) $4 \times \frac{3}{8} = \frac{12}{8}$

29) $2 \times \frac{3}{5} = \frac{6}{5}$

30) $5 \times \frac{2}{10} = \frac{10}{10}$

31) $3 \times \frac{5}{6} = \frac{15}{6}$

32) $6 \times \frac{2}{12} = \frac{12}{12}$

33) $2 \times \frac{7}{8} = \frac{14}{8}$

34) $3 \times \frac{3}{4} = \frac{9}{4}$

35) $4 \times \frac{2}{5} = \frac{8}{5}$

36) $7 \times \frac{3}{10} = \frac{21}{10}$

37) $2 \times \frac{5}{12} = \frac{10}{12}$

38) $5 \times \frac{1}{3} = \frac{5}{3}$

39) $3 \times \frac{7}{10} = \frac{21}{10}$

Fractions Times Whole Numbers: Practice (continued)

Part 4 Multiply (continued)

40) $4 \times \frac{5}{6} = \frac{\boxed{20}}{6}$

41) $6 \times \frac{3}{8} = \frac{\boxed{18}}{8}$

42) $8 \times \frac{1}{5} = \frac{\boxed{8}}{5}$

Part 5 As a mixed number No help this time. Then turn it into a mixed number.

43) $3 \times \frac{3}{4} = \frac{\boxed{9}}{4} = \boxed{2}\ \frac{\boxed{1}}{4}$

44) $2 \times \frac{4}{5} = \frac{\boxed{8}}{5} = \boxed{1}\ \frac{\boxed{3}}{5}$

45) $5 \times \frac{2}{3} = \frac{\boxed{10}}{3} = \boxed{3}\ \frac{\boxed{1}}{3}$

46) $3 \times \frac{5}{8} = \frac{\boxed{15}}{8} = \boxed{1}\ \frac{\boxed{7}}{8}$

47) $4 \times \frac{4}{6} = \frac{\boxed{16}}{6} = \boxed{2}\ \frac{\boxed{4}}{6}$

48) $3 \times \frac{7}{10} = \frac{\boxed{21}}{10} = \boxed{2}\ \frac{\boxed{1}}{10}$

49) $2 \times \frac{5}{6} = \frac{\boxed{10}}{6} = \boxed{1}\ \frac{\boxed{4}}{6}$

50) $5 \times \frac{3}{4} = \frac{\boxed{15}}{4} = \boxed{3}\ \frac{\boxed{3}}{4}$

Fractions Times Whole Numbers: Practice (continued)

Part 6 Times a fraction

51) $7 \times \frac{1}{5} = \frac{7}{5}$

52) $8 \times \frac{4}{6} = \frac{32}{6}$

53) $4 \times \frac{4}{12} = \frac{16}{12}$

54) $3 \times \frac{2}{4} = \frac{6}{4}$

55) $4 \times \frac{1}{3} = \frac{4}{3}$

56) $3 \times \frac{2}{3} = \frac{6}{3}$

57) $4 \times \frac{1}{2} = \frac{4}{2}$

58) $7 \times \frac{4}{5} = \frac{28}{5}$

59) $4 \times \frac{2}{4} = \frac{8}{4}$

60) $7 \times \frac{4}{12} = \frac{28}{12}$

61) $8 \times \frac{2}{5} = \frac{16}{5}$

62) $7 \times \frac{3}{4} = \frac{21}{4}$

Fractions Times Whole Numbers: Practice (continued)

Part 6 Times a fraction (continued)

63) $4 \times \frac{2}{6} = \frac{8}{6}$

64) $7 \times \frac{2}{3} = \frac{14}{3}$

65) $3 \times \frac{2}{8} = \frac{6}{8}$

66) $9 \times \frac{2}{4} = \frac{18}{4}$

67) $2 \times \frac{4}{8} = \frac{8}{8}$

68) $3 \times \frac{8}{10} = \frac{24}{10}$

69) $9 \times \frac{5}{8} = \frac{45}{8}$

70) $4 \times \frac{1}{6} = \frac{4}{6}$

71) $5 \times \frac{2}{6} = \frac{10}{6}$

72) $2 \times \frac{2}{3} = \frac{4}{3}$

73) $8 \times \frac{1}{12} = \frac{8}{12}$

74) $9 \times \frac{1}{5} = \frac{9}{5}$

Fractions Times Whole Numbers: Practice (continued)

Part 6 Times a fraction (continued)

75) $8 \times \frac{1}{4} = \frac{8}{4}$

76) $9 \times \frac{7}{8} = \frac{63}{8}$

77) $4 \times \frac{5}{12} = \frac{20}{12}$

78) $9 \times \frac{4}{8} = \frac{36}{8}$

79) $7 \times \frac{2}{5} = \frac{14}{5}$

80) $8 \times \frac{7}{12} = \frac{56}{12}$

81) $7 \times \frac{2}{4} = \frac{14}{4}$

82) $9 \times \frac{1}{4} = \frac{9}{4}$

83) $5 \times \frac{6}{8} = \frac{30}{8}$

84) $9 \times \frac{3}{6} = \frac{27}{6}$

85) $2 \times \frac{2}{6} = \frac{4}{6}$

86) $7 \times \frac{8}{10} = \frac{56}{10}$

Fractions Times Whole Numbers: Practice (continued)

Part 7 Remember? Problems from earlier packets.

87) $2\frac{3}{8} + 1\frac{4}{8} = 3\frac{7}{8}$

88) $\frac{11}{4} = 2\frac{3}{4}$

89) $\frac{3}{12} < \frac{1}{2}$

90) $45 \times 13 = 585$

91) $3{,}654 \div 6 = 609$

92) $\frac{5}{8} < \frac{3}{4}$

Part 8 Word problems Draw the groups.

93) Each pancake uses 3/4 cup of milk. How much milk do 2 pancakes use?

Number sentence 2 × 3/4 = 6/4

Answer 6/4 cups

Draw or show your work

94) Mia walks 2/5 of a mile to school each day. How far does she walk in 4 days?

Number sentence 4 × 2/5 = 8/5

Answer 8/5 miles

Draw or show your work

95) Each of 5 kids eats 1/8 of a pizza. How much of the pizza do they eat?

Number sentence 5 × 1/8 = 5/8

Answer 5/8 of the pizza

Draw or show your work

Fractions Times Whole Numbers: Practice (continued)

Part 8 Word problems (continued)

96) A plant grows 3/10 of an inch each week. How much does it grow in 3 weeks?

Number sentence 3 × 3/10 = 9/10

Answer 9/10 inch

Draw or show your work

97) Challenge: A recipe needs 2/3 cup of flour. Leo makes it 4 times. Is that more than 2 cups?

Number sentence 4 × 2/3 = 8/3 = 2 2/3

Answer yes

Draw or show your work

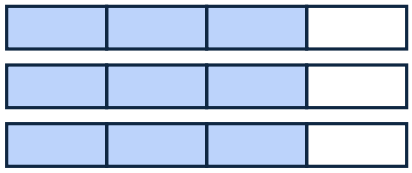
Check-In: Fractions Times Whole Numbers

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Groups of a fraction

1)



$3 \times \frac{3}{4} = \frac{9}{4}$

Part 2 Multiply

2) $5 \times \frac{1}{12} = \frac{5}{12}$

3) $3 \times \frac{7}{12} = \frac{21}{12}$

4) $5 \times \frac{3}{12} = \frac{15}{12}$

5) $2 \times \frac{2}{8} = \frac{4}{8}$

6) $5 \times \frac{2}{12} = \frac{10}{12}$

7) $5 \times \frac{7}{10} = \frac{35}{10}$

Part 3 As a mixed number

8) $2 \times \frac{7}{8} = \frac{14}{8} = 1\frac{6}{8}$

9) $4 \times \frac{2}{6} = \frac{8}{6} = 1\frac{2}{6}$